

October 8, 2026

Dear Editors of The R Journal,

We submit the article *seqbench: Anytime-Valid Sequential Benchmarking of Algorithms in R* for consideration as a contributed research article.

The article presents **seqbench**, an R package that turns the comparison of two algorithms on a benchmark into a pre-registered sequential experiment. Benchmarks in optimisation, machine learning and computational statistics are run and inspected sequentially, with adaptive budgets and early stopping, and fixed-sample tests lose their guarantees under that use. The package maintains a confidence sequence for the mean paired difference of losses that stays valid under optional stopping, adds a pre-registered margin of practical equivalence with a three-way decision rule that inherits the error control, accounts for cost and treats inconclusive outcomes as first-class results, and reports a full contract of what was assumed and consumed.

We believe the article fits The R Journal for three reasons. First, the package addresses a recurrent statistical failure in computational research with a tidyverse-native, tested and documented implementation; no package on CRAN offers this protocol. Second, the article contains a simulation study with 2,000 replications per cell, frozen before results were available, that measures the validity and cost of the package's boundaries against fixed-sample tests, an invalid repeatedly inspected test and the anytime-valid Gaussian confidence sequence of **safestats**. Third, we report the cost of distribution-free validity honestly: the Gaussian sequence with a known variance needs substantially fewer instances, and the article explains the mechanism and what a user can do about it.

Version 0.1.0 of the package is available from CRAN (<https://CRAN.R-project.org/package=seqbench>), and its documentation is at <https://castlaboratory.github.io/seqbench/>. The package has unit tests that check output values, documented examples for every user-facing function, a vignette with the full workflow and continuous integration on macOS, Windows and Linux. All experiment configurations, runner and analysis scripts are in the **experiments/** directory of the package repository (**castlaboratory/seqbench**, outside the package sources), the article source is in **paper/rjournal/** of the same repository, every table in the article is produced from archived results, and the data and scripts that rebuild the tables are included with this submission.

Sincerely,

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